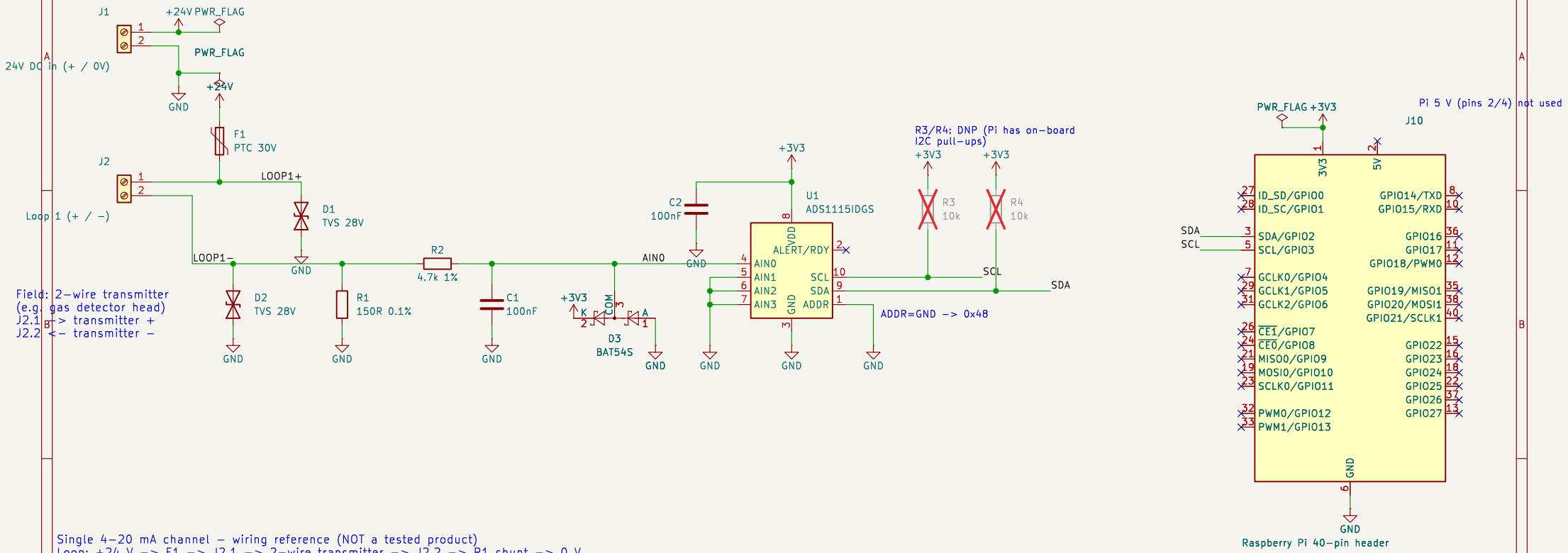


Single 4–20 mA channel – wiring reference (NOT a tested product)
Loop: +24 V -> F1 -> J2.1 -> 2-wire transmitter -> J2.2 -> R1 shunt -> 0 V
Shunt R1 = 150 R: 20 mA -> 3.00 V, 4 mA -> 0.60 V, 3.6 mA -> 0.54 V, 21 mA -> 3.15 V
  ADS1115 PGA +/-4.096 V (gain 1) at VDD 3.3 V: inputs above VDD+0.3 V are not allowed, so
  full scale is effectively 3.3 V; 21 mA still leaves 150 mV headroom.
  20 mA: P = I^2 R = 60 mW (use >= 0.25 W, 0.1 %, 25 ppm/K or better).
  Loop burden at 20 mA: 3.0 V across R1 (+ PTC/wiring); transmitter sees the rest of 24 V.
R2 + C1: fc = 1/(2 pi 4.7k 100n) ~ 340 Hz anti-alias / RF filter; R2 limits clamp current:
  worst case (transmitter shorted, 24 V on LOOP−): (24 − 3.6) V / 4.7k ~ 4.3 mA < 10 mA abs-max.
D3 (BAT54S) clamps AIN0 to the 3.3 V rail / GND; D1, D2 absorb field surges (choose standoff >= supply).
F1 PTC: with a shorted transmitter R1 sees 24 V / 150 R = 160 mA (3.8 W) until F1 trips – pick a
  PTC/fuse that opens fast enough for the shunt's pulse rating (see docs/wiring-single-channel.md).
3-wire transmitter: +24 V -> sensor V+, sensor 0 V -> GND (0 V), sensor OUT -> J2.2. J2.1 not used.
4-wire / self-powered: connect its loop output + to J2.2 and its − to GND only if its output
  is isolated or shares this 0 V; otherwise use a loop isolator.
Common ground: the 24 V supply 0 V, ADS1115 GND and Pi GND are one net.
Recommended: power U1 from a GPIO-switchable 3.3 V rail so the I2C bus can be recovered after a brown-out.
R3/R4 are DNP: the Raspberry Pi already has pull-ups on GPIO2/GPIO3.

mqttpi-4-20ma – MIT licensed
Monitoring/telemetry only – not a certified gas-detection controller
Untested reference design – verify against datasheets before building

Sheet: /
File: ads1115-4-20ma-single.kicad_sch

Title: ADS1115 + one 4–20 mA loop (2-wire) – reference wiring

Size: A4	Date: 2026-10-27	**Rev: 0.1**
KiCad E.D.A. 9.0.2		Id: 1/1